Research Article

Bio-Inspired Multifunctional 3D Printed Protective Structures

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ABSTRACT: Modern body protective structures systems, while effective against threats, often lack the ability to detect internal micro-cracking and degradation caused by repeated use. To address this issue, this study introduces a new self-sensing protective structures architecture that combines bio-inspired designs with piezoresistive materials, specifically conductive carbon-black polylactic acid (CB-PLA) and carbon-black thermoplastic polyurethane (CB-TPU). Lattices were designed to replicate the features of the deep-sea glass sponge (*Euplectella aspergillum*) and the impact-resistant dactyl club of the mantis shrimp (*Odontodactylus scyllarus*).

Models of these bio-inspired structures were designed using CAD software, fabricated using fused deposition modeling (FDM), and subjected to compression testing while changes in electrical resistance were measured. Analysis of the experimental results shows a correlation between compressive loading and changes in resistance before model failure under excessive strain. These results support the potential of combining bio-inspired designs with conductive materials to develop the creation of lightweight, self-sensing protective structures systems.

KEYWORDS: *Bioinspired Design, 3D Printing, Multifunctional Composites, Protective Gear.*

INTRODUCTION

The majority of body protective structures on the market today is made of synthetic materials such as Dyneema and Kevlar. While highly effective against bullets, these materials are less effective against large impacts and present significant structural health monitoring (SHM) issues [1]. Internal micro-cracking from repeated impacts can go unnoticed without embedded sensors or methods such as infrared thermography and acoustic emission monitoring, which require specialized, costly, and large external testing equipment, making continuous or real-time monitoring impractical in field settings. Because this damage may occur without visible surface signs, determining when protective structures have been compromised and should be repaired or replaced remains difficult.

Despite manufacturing advancements, it remains difficult to create smart materials that combine bio-inspired features with reliable sensing [2]. Current models fall short because they fail to account for how internal electrical networks shift under pressure or how electrical signals distort when the material is permanently damaged. In particular, the relationship between a material's piezoresistive response and its ability to absorb and dissipate impact energy remains poorly understood. As a result, improving these complex structures still relies heavily on trial and error rather than a clear understanding of their microscopic behavior.

In light of these limitations, this study proposes 3D-printed, self-sensing, and customizable smart protective structures. It focuses on the design and additive manufacturing of the bio-inspired lattice structures using conductive carbon-black polylactic acid (CB-PLA) and carbon-black thermoplastic polyurethane (CB-TPU). These structures are inspired by the deep-sea glass sponge (*Euplectella aspergillum*) [3,5] and the mantis shrimp dactyl club (*Odontodactylus scyllarus*) [4], both of which have evolved highly optimized structures for distributing force, absorbing energy, and resisting fracture. The glass sponge is particularly useful as a model because its hierarchical lattice can withstand substantial loading while using relatively little material.

Additive manufacturing makes these bio-inspired geometries practical to fabricate while allowing rapid prototyping and future customization to the wearer. By printing lattice structures with CB-PLA, the structures gain piezoresistive capabilities [6]. Under stress, cracks form, some of which are invisible to the naked eye. This alters the conductive network formed by the carbon particles, producing measurable shifts in electrical resistance [7].

To systematically optimize this behavior, the study evaluates eight line-thickness variations and five line-spacing variations, comparing how changes in geometry affect both mechanical performance and electrical response. By testing these variations under compressive loads, this research examines the relationship between electrical resistance, deformation, and failure mechanisms, with the aim of enabling real-time structural health monitoring built directly into the protective structures without relying on external sensors.

Ultimately, the goal is not simply to create protective structures that sense deformation, but protective structures whose electrical response can reveal how much damage it has accumulated and whether its structural integrity has been compromised.

METHODS

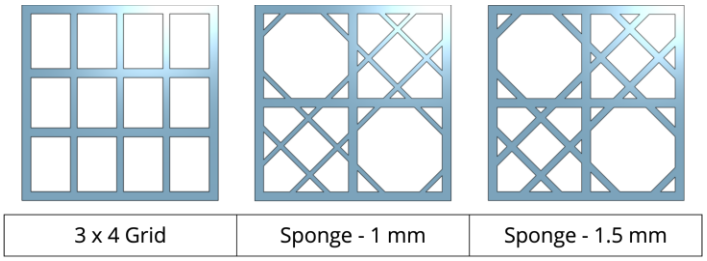


Figure 1: CAD views of the 3 × 4 control grid and glass sponge-inspired lattices with diagonal line thicknesses of 1.0 and 1.5 mm.

Design and Modeling

All digital prototypes were created using Onshape, a web-based computer-aided design platform. This platform was chosen because it has a workspace and multi-part studio capabilities. A standard control model was made to create a baseline for resilience and electrical resistance. This model, as shown in Figure 1, has a 3 x 4 grid. The overall size is 44 mm x 44 mm x 5 mm. The walls have a thickness of 2.0 mm.

After making the control model, the first inspired design was created. This design was based on the force-distributing microscale skeletal lattice of the deep-sea glass sponge [5]. This is shown in Figure 2. This design retained the control model's outer dimensions of 44 mm x 44 mm x 5 mm, allowing us to isolate the effect of diagonal cross-bracing on load distribution. To test versions of the design, global variables were used in Onshape. This allowed changes to parameters such as strut thickness and cell pitch. These changes could be made without rebuilding the geometry from scratch.

To systematically evaluate the design space, a total of eight line-thickness variations and five line-spacing variations were investigated. Specifically, line spacing was adjusted relative to the baseline nominal value of 8.721 mm by applying offsets ranging from -1.0 mm to +1.0 mm in 0.5 mm increments. These offsets correspond to absolute diagonal line spacings of 7.721mm (-1.0 mm offset), 8.221 mm -0.5 mm offset, 8.721 mm (regular baseline), 9.221 mm (+0.5 mm offset), and 9.721 mm (+1.0 mm offset).

High geometric density caused some problems with regeneration and surface export. These problems were fixed by simplifying constraints and using Boolean unions. This setup made it possible to create variations for multiple rounds of testing. This is shown in Figure 1.

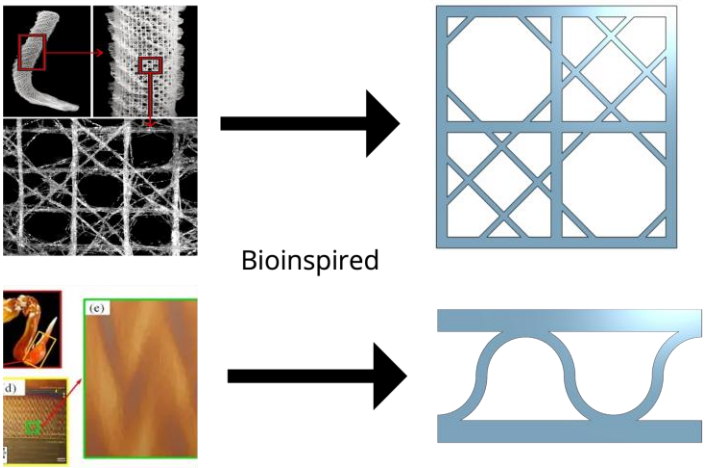


Figure 2: Natural structures and their corresponding bio-inspired CAD designs. The glass sponge lattice inspired the rigid lattice model, and the mantis shrimp's dactyl club inspired the arched model.

The second inspired design was based on the impact-resistant dactyl club of the mantis shrimp. This is shown in Figure 2. These models were made to test energy dissipation through deformation. This is different from structural failure. The models have an array of arches or " arches". The arches were created using sweep tools, as shown in Figure 3. Fixing non-manifold edge errors along the arch boundaries made the geometry watertight. This was needed for analysis. Because of the surface shape, the models ended up with a size of 25 mm by 15 mm by 5 mm. Changing the number, pitch, and height of the arches made it possible to test how bump density affects performance and strain recovery.

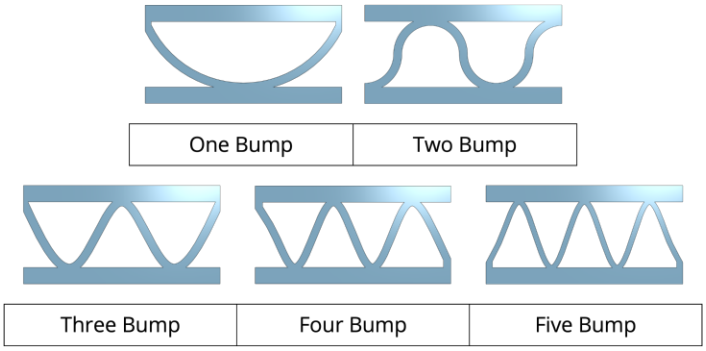


Figure 3: CAD elevation views of mantis shrimp-inspired models with one to five load-bearing arches, or " arches." Overall dimensions were held constant across the designs.

Materials and Fabrication

Fused deposition modeling (FDM) 3D printing was used to fabricate the glass sponge and shrimp dactyl club models [8]. We sliced the lattice models using Bambu Studio (slicing software). We created a base geometry by sketching a simplified square design on the bottom plane in Onshape (CAD platform), inspired by previous papers [3]. Key geometric parameters, including the outer wall thickness, depth, corner spacing, and scaling, were controlled to maintain consistent dimensions between samples while allowing individual design variables to be modified. To manufacture each model, we uploaded STL files to the Bambu Lab A1 and H2C 3D printers. Printing settings included a 0.4 mm nozzle, 15% infill, and a "gyroid" infill preset, with print orientation as an individually tested variable for both horizontal and vertical printing. The filaments used were the conductive CB-PLA and CB-TPU. The printing procedure for the mantis shrimp models differed because CB-TPU required additional handling [9]. Due to CB-TPU's flexible nature, it wasn't loaded through a PTFE tube and required extra setup. Filament was kept dry, and adjustments to the idler screws on the Bambu Lab H2C printer were made, loosening them by 1.5 complete turns. Then, filament was manually fed into the extruder by pinching its tip. Through these methodical procedures, the CB-TPU filament could be loaded reliably.

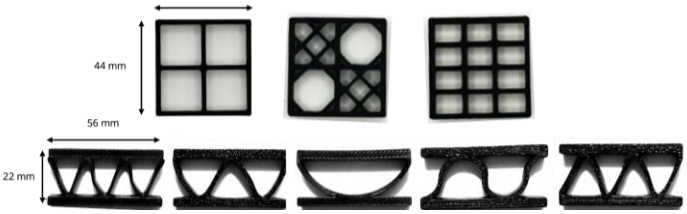


Figure 4: Representative 3D-printed grid, glass sponge-inspired, and mantis shrimp-inspired models. The models illustrate the lattice and arched geometries fabricated for compression testing.

During the process of printing the lattice models, a few issues occurred, including filament stringing and difficulty printing small features. Most printing errors occurred when printing smaller lattice features, leading to filament stringing or models printed without the desired details. In the case of the 0.75 mm thickness model, it was unable to print the majority of the lattice design, only being able to print the surrounding edges as shown in Figure 4. This resulted in a simple 2-by-2 grid. Despite the printing challenges encountered, every other model iteration printed smoothly, leading to successful production of consistent, functional samples.

The final prints were approximately 44 x 44 x 5 mm, with a line thickness of 1.0 mm and a line spacing of 8.721 mm, although this was changed during testing to see if results improved. It is noted that conductive CB-PLA was selected for its higher electrical conductivity and structural rigidity, while CB-TPU's flexibility and elastic deformation were considered for shock and vibration absorption [10].

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Figure 5: Compression testing setup. A hand-cranked stand and an Mxmoonfree 500 N force gauge applied compressive force through a cylindrical tip with a diameter of 16 mm. A digital scale measured displacement, and an HT118E True RMS digital multimeter connected with alligator clip leads measured electrical resistance.

Data Analysis and Statistics

For the purpose of evaluating the models, they were each mechanically tested using a hand-operated compression machine paired with a force gauge, which recorded applied force and displacement, while a digital multimeter simultaneously measured electrical resistance across the sample, as shown in Figure 5 [11]. Running these two measurements in parallel allowed the correlation between mechanical loading and resistive response to be tracked directly as each model was compressed to failure. Each model was tested three times, and the results were averaged to provide a representative value and reduce the influence of individual outliers.

Testing was carried out across multiple rounds, with each successive round building on the results of the previous one to progressively refine the design toward a finalized configuration.

The data collection process was not without setbacks. The compression machine's 500 N maximum force limit, combined with its hand-operated actuation, meant the rate of force application could not be perfectly controlled across trials, and models able to withstand the full 500 N force could not be tested to failure. On the data-handling side, our testing software unexpectedly deleted several completed test datasets, and additional difficulties working with Google Sheets and other graphing tools slowed the process of organizing and visualizing results. Resistance readings also proved inconsistent across identical prints, as a result of minor manufacturing imperfections.

RESULTS AND DISCUSSION

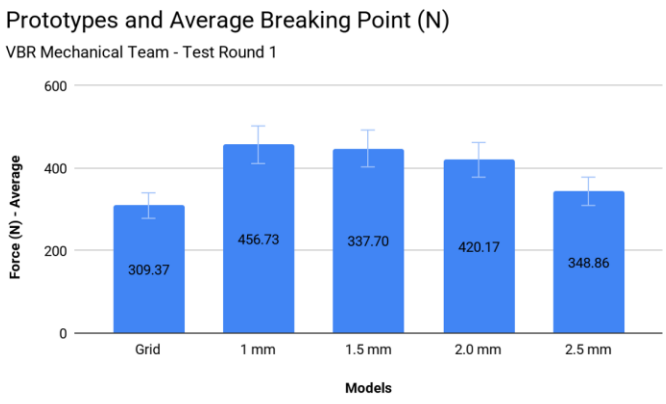


Figure 6: Average maximum force withstood by the control grid and glass sponge-inspired models with different diagonal line thicknesses, based on three trials per model. The 1.0 mm model had the highest average maximum force (456.733 N), and the control grid had the lowest (309.367 N).

Round 1 of Testing

Finding the ideal cross-line thickness for the glass sponge lattice structure was the main goal of the first evaluation. Across thicknesses of, 1.0 mm, 1.5 mm, 2.0 mm, and 2.5 mm, the 1.0 mm model outperformed the 3 x 4 grid, which failed at 309.37 N, and showed the strongest mechanical resilience, as seen in Figure 6, withstanding an average peak force of 456.73 N before complete failure.

According to Figure 7, the 1.0 mm arrangement also offered the highest change in resistance (ΔΩ) under load. These results

suggest that the 1.0 mm geometry provides the best balance, possibly reflecting differences in the conductive carbon-black network within the PLA [6]. The electrical disruption brought on by localized micro-cracking was most likely hidden by thicker structural lines (e.g., 2.5 mm), which also decreased the model's self-sensing sensitivity [11].

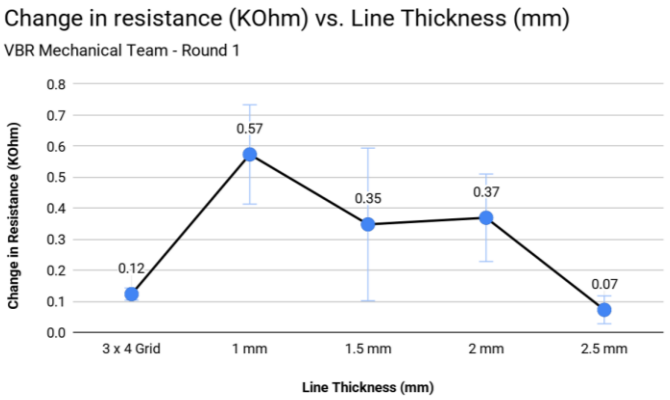


Figure 7: Change in electrical resistance under compression for the control grid and glass sponge-inspired models with different diagonal line thicknesses. The 1.0 mm model showed the largest resistance change.

Round 1.5 of Testing

To further evaluate whether 1.0 mm was the optimal thickness, a secondary testing phase was conducted that introduced increments of 0.75 mm, 0.9 mm, and 1.25 mm. In the testing, the 1.0 mm model maintained both its load-bearing capacity and its high change in resistance compared to other models. The performance dropped off significantly at lower thicknesses, especially in the 0.75 mm model, which, when printed with a standard 0.4 mm nozzle, was unable to print properly. This compromised the structural lattice, resulting in the lowest average failure point at 165.7 N as shown in Figure 8, and rendering the model ineffective for our use cases.

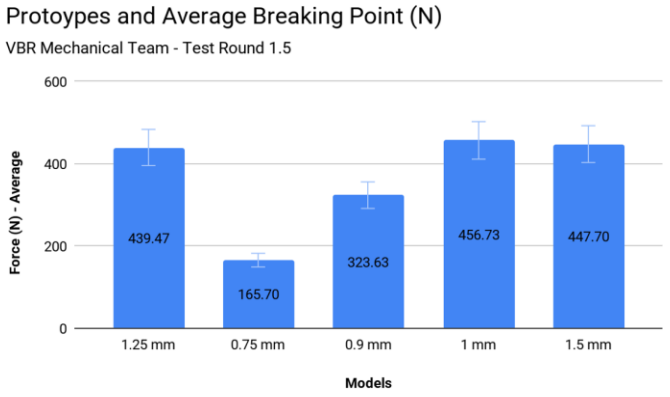


Figure 8: Average failure force across three trials for each glass sponge-inspired model evaluated in the secondary thickness-testing phase. The 1.0 mm model had the highest average failure force (456.73 N), while the partially printed 0.75 mm model had the lowest (165.7 N).

Round 2 of Testing

After determining the optimal line thickness (1.0 mm), line spacing was adjusted across five variations, spanning offsets from -1.0 mm to +1.0 mm relative to the 8.721 mm baseline (corresponding to absolute spacings of 7.721 mm, 8.221 mm, 8.721 mm, 9.221 mm, and 9.721 mm). After testing, results showed that the +0.5 mm offset (9.221 mm absolute spacing) exhibited the best overall mechanical performance, withstanding an average peak force of 493.73 N as shown in Figure 9 and distributing the compressive force better than the other models.

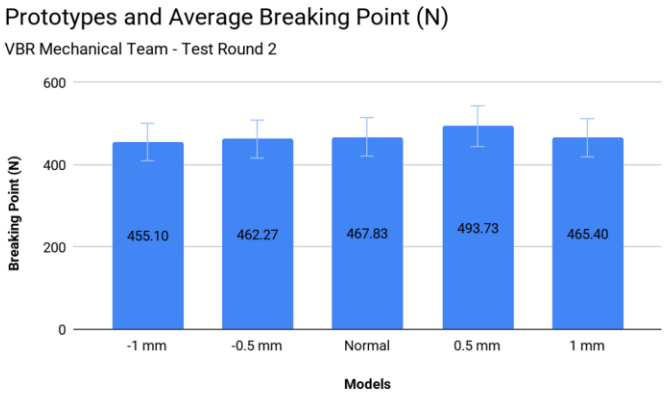


Figure 9: Average failure force for glass sponge-inspired models with line-spacing offsets from −1.0 to +1.0 mm in 0.5 mm increments. The +0.5 mm model had the highest average failure force (493.73 N).

Notably, the +1.0 mm model had increased electrical sensitivity compared to the other models—reaching a peak of 1.098 kΩ as shown in Figure 10, yet this came at the cost of structural stability, as larger spacings created more weak points. Because of this, the +1.0 mm model only withstood 465.40 N.

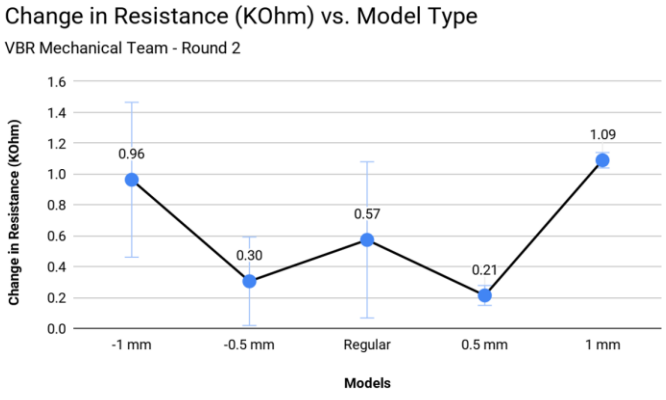


Figure 10: Average change in electrical resistance under compression for glass sponge-inspired models with different line-spacing offsets. The +1.0 mm model showed the largest average change, and the +0.5 mm model showed the smallest.

Round 3 of Testing

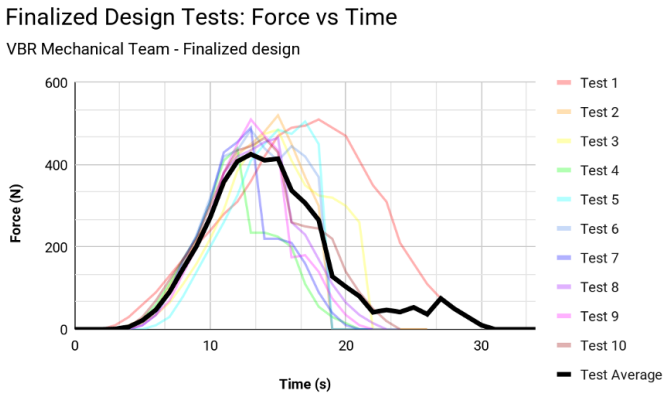


Figure 11: Force versus time for ten compression tests of the finalized glass sponge-inspired design. Colored curves represent individual tests, and the black curve represents the average response.

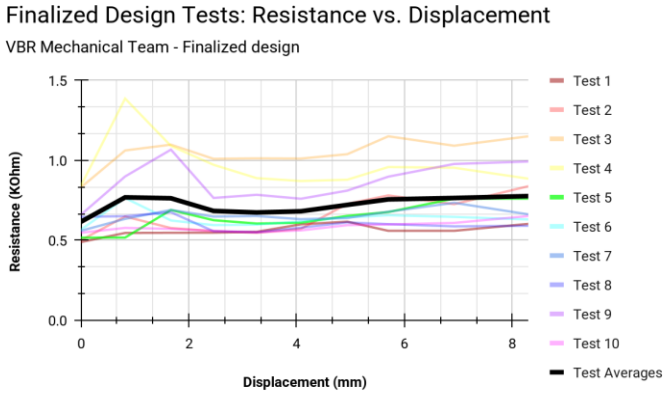


Figure 12: Electrical resistance versus displacement for ten compression tests of the finalized glass sponge-inspired design. Colored curves represent individual tests, and the black curve represents the average response.

The finalized glass sponge design integrated the 1.0 mm line thickness with a +0.5 mm spacing and was then subjected to ten tests alongside FEA. Ten replicate tests were conducted to evaluate the repeatability of the finalized design and reduce the influence of experimental variability. As shown in Figure 11, the finalized design was able to withstand a peak force of 425.50 N, and as shown in Figure 12, reached an average peak of 0.76 kΩ.

Mantis Shrimp TPU Testing

Testing on the flexible CB-TPU models shifted our focus from brittle fracture sensing to elastic energy absorption [10]. Models featuring varying top-cover topologies (one to five continuous arches) were compressed to evaluate their load-bearing efficiency under elastic deformation.

In our testing, the two-bump model proved to perform substantially better than the other models, withstanding the highest peak force of 131.67 N, as shown in Figure 13. Its wider arches may have distributed the applied load more effectively. Conversely, increasing the number of arches did not increase strength; the four-bump model performed the worst, reaching the lowest peak force of 92.33 N. The higher frequency of smaller, tighter arches introduced structural instability, causing the material to undergo localized buckling rather than uniform compression.

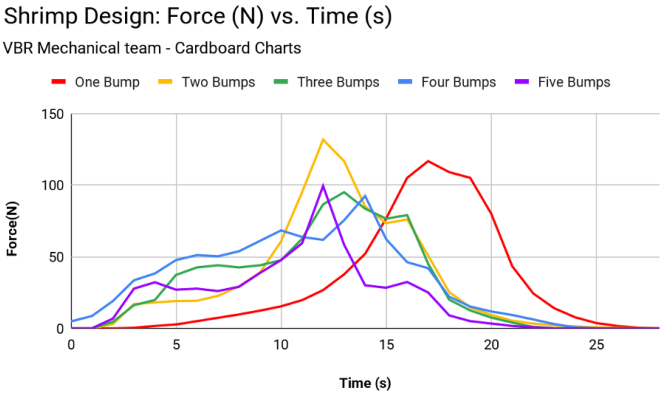


Figure 13: Force versus time for mantis shrimp-inspired CB-TPU models with one to five arches. The two-bump model reached the highest peak force (approximately 132 N).

CONCLUSION

We have shown that 3D printing can integrate bio-inspired designs with conductive materials to create self-sensing structures for potential protective structures. By mapping changes in electrical resistance to compressive loading, we established a measurable relationship between electrical and mechanical behavior. This piezoresistive mechanism allows the structural material itself to act as a diagnostic sensor, potentially reducing reliance on external equipment for non-destructive protective structures evaluation methods [1].

Combined mechanical and electrical evaluations identified promising design parameters. For rigid components, the deep-sea glass sponge lattice printed in CB-PLA provided a useful balance of structural resilience and electrical response with a 1.0 mm cross-line thickness and a +0.5 mm line-spacing offset. For flexible layers, the mantis shrimp-inspired two-bump CB-TPU design showed the best compressive load performance among the models tested, with less localized buckling.

Future research will evaluate alternative materials, such as PETG and ABS, to determine their effects on structural toughness and electrical sensitivity. Automated, motorized loading equipment could also improve testing precision and reduce human error. Combining bio-inspired geometries with 3D-printable conductive materials offers a promising approach for next-generation protective systems, including military body protective structures, sports safety equipment, and medical prosthetics [10, 12-14].

ACKNOWLEDGEMENTS

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AUTHOR CONTRIBUTIONS

All authors contributed to conceptualization, methodology, 3D printing, testing, validation, investigation, writing of the original draft, and review and editing.

ETHICS STATEMENT

Not applicable

DATA AVAILABILITY

All data supporting the findings of this study are presented in the paper.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

REFERENCES

[1] Colorado, Henry A., Carlos A. Cardenas, Elkin I. Gutierrez-Velazquez, Juan P. Escobedo, and Sergio Neves Monteiro. "Additive manufacturing in armor and military applications: research, materials, processing technologies, perspectives, and challenges." Journal of Materials Research and Technology 27 (2023): 3900-3913.

[2] Wei, Jingjiang, Fei Pan, Hang Ping, Kun Yang, Yanqing Wang, Qingyuan Wang, and Zhengyi Fu. "Bioinspired additive manufacturing of hierarchical materials: From biostructures to functions." Research 6 (2023): 0164.

[3] Fernandes, Matheus C., Joanna Aizenberg, James C. Weaver, and Katia Bertoldi. "Mechanically robust lattices inspired by deep-sea glass sponges." Nature Materials 20, no. 2 (2021): 237-241.

[4] Behera, Rohit Pratyush, and Hortense Le Ferrand. "Impact-resistant materials inspired by the mantis shrimp's dactyl club." Matter 4, no. 9 (2021): 2831-2849.

[5] Robson Brown, K, D Bacheva, and R S Trask. 2019. “The Structural Efficiency of the Sea Sponge Euplectella Aspergillum Skeleton: Bio-Inspiration for 3D Printed Architectures.” Journal of the Royal Society, Interface. England. PubMed Central.

[6] Peng, Ke, Tianyu Yu, Pan Wu, and Mingjun Chen. "Piezoresistive porous composites with triply periodic minimal surface structures prepared by self-resistance electric heating and 3D printing." Sensors 24, no. 7 (2024): 2184.

[7] Stano, Gianni, Antonio Pavone, Md Abu Jafor, Khaled Matalgah, Gianluca Percoco, and Trevor J. Fleck. "Enhancing the sensitivity of 3D printed sensors via ironing and void reduction." Virtual and Physical Prototyping 19, no. 1 (2024): e2331153.

[8] Velasco-Hogan, Audrey, Jun Xu, and Marc A. Meyers. "Additive manufacturing as a method to design and optimize bioinspired structures." Advanced Materials 30, no. 52 (2018): 1800940.

[9] Bandyopadhyay, Amit, Susmita Bose, and Suman Das. 2015. “3D Printing of Biomaterials.” MRS Bulletin 40 (2): 108–15.

[10] Lee, Hyojeong, Ran-I. Eom, and Yejin Lee. "Evaluation of the mechanical properties of porous thermoplastic polyurethane obtained by 3D printing for protective gear." Advances in Materials Science and Engineering 2019, no. 1 (2019): 5838361.

[11] He, Liu, Peiren Wang, Junhui Yang, Kaoyi Fan, Hanqiang Zhang, Luyan Zhang, Mingxing Jiang et al. "Smart lattice structures with self-sensing functionalities via hybrid additive manufacturing technology." Micromachines 15, no. 1 (2023).

[12] Jammalamadaka, Udayabhanu, and Karthik Tappa. 2018. “Recent Advances in Biomaterials for 3D Printing and Tissue Engineering.” Journal of Functional Biomaterials 9 (1): 22.

[13] Yang, Yang, Xuan Song, Xiangjia Li, Zeyu Chen, Chi Zhou, Qifa Zhou, and Yong Chen. 2018. “Recent Progress in Biomimetic Additive Manufacturing Technology: From Materials to Functional Structures.” Advanced Materials 30 (36): 1706539.

[16] Yang, Yang, Xiangjia Li, Ming Chu, Haofan Sun, Jie Jin, Kunhao Yu, Qiming Wang, Qifa Zhou, and Yong Chen. 2019. “Electrically Assisted 3D Printing of Nacre-Inspired Structures with Self-Sensing Capability.” Science Advances 5 (4): eaau9490.

[14] He, Qingqing, Yushun Zeng, Laiming Jiang, Ziyu Wang, Gengxi Lu, Haochen Kang, Pei Li, Brandon Bethers, Shengwei Feng, Lizhi Sun, Peter Sun, Chen Gong, Jie Jin, Yue Hou, Runjian Jiang, Wenwu Xu, Eugene Olevsky, and Yang Yang. 2023. “Growing Recyclable and Healable Piezoelectric Composites in 3D Printed Bioinspired Structure for Protective Wearable Sensor.” Nature Communications 14: 6477.